More Context, Less Accuracy: A Pre-Registered Negative Result for a Byte-Level Decider

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Abstract

A decider replaces generation with scoring a finite candidate set. The architectural promise is that a small local model can match a hosted service at a fraction of the cost, but the promise is usually tested on a different row set, a different prompt and a different notion of "the same information". We run the comparison the promise needs. On all 3,080 official Banking77 test rows, with the exact 24 retrieved examples and the frozen per-row correctness vector of a 92.40% Jev run, a 113.47M-parameter byte-level decider reaches 89.03% (95% Wilson 87.87–90.08), a gap of 3.38 points. The pre-registered bar is missed (exact McNemar $p = 3.99 \times 10^{-9}$; paired 95% lower bound -4.32 points). The frozen examples contain 25 test messages (255 rows retrieve at least one); a decontaminated rerun scores 89.06%, so the gap is not an artefact of that overlap.

The negative result has a located mechanism. Mean pooling dilutes the customer message into the retrieved examples: the same encoder falls from 88.38% message-only to 59.12% with examples at its native 1,020-byte window, and to 22.44% when all 24 examples are packed at 5,120 bytes. Replacing mean pooling with a message-query attention pool recovers +29.90 points on identical contexts (+28.41 lower bound), which identifies the pooling operation rather than the examples as the immediate cause. The fix is real but not sufficient. The query head with examples is only +0.65 points above the old head's message-only arm (not significant; $p = 2.60 \times 10^{-1}$), and a joint 5,120-byte fine-tune that fits all 24 examples reaches 87.11%, below both the native-window query head and the message-only head.

The comparator's reconstructed JSON prompt is 10,683 bytes in the median, 2.09× the largest context we test, and contains the 77 category definitions that no Azurite arm receives. The comparison is therefore not information-matched. We publish the full per-row records, the hashes and a generator that recomputes every number in this paper.

1 Introduction

A decider constrains a model's output to a declared candidate set.[1] Instead of generating a label token by token, it scores every candidate and returns the argmax; an out-of-bank answer is impossible by construction. TypeSafe's Jev markets this interface as a "System One" model: all outputs are generated in a single query, the possible outputs are defined in advance, and the service is priced per input token (Almeida, 2026). The interface is attractive for classification, routing and extraction because it converts an open-ended generation problem into a finite comparison.

The commercial argument is comparative: a compact local decider should be faster, smaller and cheaper than a hosted decider at equal accuracy. The accuracy clause is the hard one. A hosted decider can be given a long prompt with category definitions and retrieved labelled examples; a local decider is limited by its context window, its training distribution and its pooling operation. If the local model fails, the failure could be the architecture, the information, or the interface. A comparison that changes all three at once cannot say which.

This paper isolates one of those changes. We take a frozen third-party run of Jev 1.13 on the complete official Banking77 test set: 2,846 of 3,080 rows correct (92.40%), with the exact 24 BM25-retrieved labelled examples and all 77 category definitions supplied for every row (simonmesmith, 2026). We give Azurite, a 113.47M-parameter byte-level encoder with a decider head, the same rows, the same gold labels, the same 24 examples and the same frozen Jev correctness vector. We withhold the category definitions, so Azu-

[1] This manuscript is formatted for EMNLP 2026 using the official ACL style files in review mode.

rite's information budget is strictly smaller; that asymmetry is stated before the result and favours Jev. We then vary the head and the context budget under separate pre-registrations.

The headline is negative. The best arm reaches 89.03%, 3.38 points below Jev, and the pre-registered P0 rule is missed. We report that plainly. The more useful result is why. Mean pooling, which is the default for sentence-level classification, averages every non-padding token. A context with a short customer message and several retrieved examples is mostly examples, so the pooled vector stops being a representation of the message. The same encoder therefore loses -33.28 points when examples are added at its native window, and collapses to 22.44% when all 24 examples are packed at 5,120 bytes. A message-query attention pool removes that dilution and recovers +29.90 points on identical contexts. That is a clean mechanism result: the examples were readable, but the read-out was averaging them away.

Removing the pooling bottleneck does not recover the accuracy. The query head with examples is statistically indistinguishable from the old head with no examples at all (+0.65 points, $p = 2.60 \times 10^{-1}$). Extending the context to 5,120 bytes so that all 24 examples fit makes things worse: the joint fine-tune reaches 87.11%, -1.92 points below the native query head and -1.27 points below the old message-only head. The example budget is not the missing variable. The comparator's reconstructed JSON prompt is 10,683 bytes in the median and 7,801 of those bytes are the 77 definitions; our largest context is 47.9% of the median prompt and contains no definitions. The experiment therefore bounds the claim it can make. It shows that this decoder, with this information budget and this pooling, cannot reach the bar. It does not show that a decider with information parity cannot.

Contributions.

1. A pre-registered comparison on the full 3,080-row Banking77 test set, using the exact per-row examples and frozen correctness vector of a public 92.40% Jev run, with paired Wilson, McNemar and bootstrap inference.

2. A located mechanism for the failure: mean-pool dilution, isolated by a same-context head control and a message-only control, and corroborated by prediction concentration on the stored per-row predictions.

3. A bound on the example-budget explanation: all 24 examples fit at 5,120 bytes, yet accuracy falls; the comparator prompt is 2.09× larger and includes definitions. The remaining gap is an information-and-scale question, not another head.

4. A reproducibility artifact: every number, table and figure is emitted from the stored summaries by one generator, with a reconciliation record that also checks the overlapping thesis values.

Scope. This is a study of a text-only, byte-level decider at a 5,120-byte context. It makes no claim about multimodal models or about long-context models in the modern sense: 5,120 bytes is four times the native window but orders of magnitude below the contexts used in long-context evaluations (Liu et al., 2024; Hsieh et al., 2024). The negative result is correspondingly narrow, and we state the boundary explicitly rather than extrapolating from it.

2 Related Work

Deciders and constrained output. Jev is the closest comparator: a hosted service that returns one of a declared set of outputs and reports per-token pricing and latency (Almeida, 2026). The public Banking77 run we use supplies retrieved examples and category definitions and scores all 3,080 test rows (simonmesmith, 2026). Constrained decoding and candidate scoring are older ideas; what is new in the decider framing is the claim that a finite, declared output space makes a small model competitive with open-ended generation. Our experiment tests the accuracy half of that claim under a matched row set, not the pricing half.

Retrieval and in-context examples. Retrieval-augmented models condition on passages or demonstrations selected at inference time (Karpukhin et al., 2020; Lewis et al., 2020). Few-shot prompting shows that demonstrations can change a model's behaviour substantially (Brown et al., 2020), but the effect is not simply "more examples, more accuracy": demonstrations can supply the label space and the format while

the input-label mapping matters less than expected (Min et al., 2022), and many-shot regimes can behave differently from few-shot ones (Agarwal et al., 2024). Our result is consistent with that literature but is a different failure: the examples are not ignored, they are averaged into the representation until the message is no longer recoverable.

Byte-level encoders. Byte-level models remove the tokeniser and make the input distribution uniform over bytes (Xue et al., 2022; Clark et al., 2022). That is attractive for a decider because the candidate set and the input share one alphabet, and because arbitrary labels can be scored without a tokenisation mismatch. The cost is a longer sequence for the same text and a greater sensitivity to context length. Azurite is a masked byte-level encoder in this line; the head is a candidate scorer rather than a generator.

Evaluation and inference. A single accuracy on a single draw is a noisy estimate. We use Wilson score intervals (Wilson, 1927), exact McNemar tests on paired correctness (McNemar, 1947) and a paired bootstrap lower bound, following the recommendation that classifier comparisons be paired rather than treated as independent samples (Dietterich, 1998). The protocols were written and hashed before evaluation; amendments are recorded and limited to evaluator or schedule changes. This is a small methodological contribution, but it is what makes a negative result interpretable.

3 Protocol

3.1 Task and frozen comparator

Banking77 is a fine-grained intent classification task with 77 banking intents (Casanueva et al., 2020). We use the official test split of 3,080 rows and the 4,000 training rows. The public Jev run supplies, for every test row, the customer message, the gold label, the Jev prediction, a binary Jev-correctness bit, and exactly 24 BM25-retrieved labelled training examples with all 77 category definitions (simonmesmith, 2026). We freeze that record before any Azurite forward pass: the row order, the examples, the gold labels and the Jev correctness vector are identical in every arm. The frozen vector has 2,846 correct rows, so the pre-registered P0 bar is 92.40%.

BM25 is the retriever used by the public harness (Robertson and Zaragoza, 2009). The retrieval is not re-run: we take the stored example ids and messages exactly as the public run recorded them. The public run describes the examples as training-split retrievals, but the stored record does not fully support that description. Under normalised text (NFKC, case-folding, punctuation collapsed), 25 of the 9,993 support messages are official test rows (6 exact; 0.25% of the bank), and all carry the same gold label. They occur in the frozen example sets of 255 of the 3,080 rows (8.28%), as 283 instances: 25 self-retrievals and 258 cross-row retrievals. The contamination is not uniform: 2,825 rows retrieve no test text. The frozen set therefore contradicts the comparator's description of its examples as training-split-only retrievals.

The original product path over the 9,993-row bank is not an independent corroboration: it produced the same prediction on all 3,080 rows and the same context digest on 2,971 of them. Removing all 25 test messages from the support bank and re-running gives 89.06% (2,743/3,080; 95% Wilson 87.91–90.11), a change of +0.03 points (2 wins, 1 loss; exact McNemar p = 1.00). The 2,825 unaffected rows score 88.88% (2,511/2,825; 95% Wilson 87.67–89.99). The overlap does not explain the gap, but 89.03% is not a clean held-out estimate and must not be presented as one.

3.2 Model and arms

Azurite is a 113.47M-parameter masked byte-level encoder (Xue et al., 2022; Clark et al., 2022), built from transformer blocks (Vaswani et al., 2017) and trained with a masked objective (Devlin et al., 2019). The encoder has a native window of 1,024 bytes. Two heads are compared.

Old head. A mean-pool over every non-padding token, followed by a linear projection to the 77 candidate embeddings. This is the standard sentence-classification read-out and the head used by the earlier suite. It is trained on 4,000 rows with the message only.

Query head. A masked attention pool whose query is a linear map of the mean-pooled customer message. Candidate embeddings are shared with the old head's projection. The head is trained on the same 4,000 rows, but the contexts contain retrieved examples; the message-query is what keeps the message from being averaged away.

Table 1 lists the arms. The old head is evaluated at 1,020 bytes (message only and the exam-

ples that fit) and at 5,120 bytes with all 24 examples. The query head is evaluated at 1,020 bytes (message only and examples) and at 5,120 bytes with all 24 examples. The 5,120-byte query arm is a joint fine-tune: the encoder and head are trained together for 3,000 steps at batch 8, which is 122,880,000 byte positions and 1.69 GPU-hours on one RTX PRO 6000. The original two-stage length-adaptation schedule was replaced by this joint schedule in a hashed amendment before any 5,120-byte accuracy existed; the amendment changes the schedule, not the question, rows, examples or decision rule. The joint schedule means the arm cannot separate length from head training, and we treat it that way.

Decision rule. The primary arm is fixed in advance. The P0 bar is Jev's point estimate, 92.40%. The arm *exceeds* P0 if its point estimate is above the bar and the exact two-sided McNemar test against the frozen Jev vector has $p < 0.05$. It *meets* P0 if its point estimate is at least the bar. It *misses* P0 otherwise. Non-inferiority is reported separately: the 5th percentile of 10,000 paired bootstrap resamples (seed 0) must exceed -2.0 points. No early stopping, checkpoint selection or best-of-steps choice is used; the final step is evaluated.

3.3 Provenance

The query protocol is hashed as `bde97e74f2f8`; the 5,120-byte protocol as `c59b5e08d916` and its schedule amendment as `7993177d769a`. The frozen inputs hash to `8dccd8c89886`; the old head to `42f1ed1c8471`, the query head to `7992c97362b3`, and the 5,120-byte head to `e7455fc462e9`. The 5,120-byte run starts from the 1,024-position checkpoint `5294fd685615`. Table 3 in the appendix lists the same values, and `paper/RECONCILIATION.md` records where each paper number comes from.

Every arm writes one JSONL record per row with the row id, the gold index, the predicted index, the correctness bit, the context hash, the context byte count and the number of effective examples. The evaluation summary is generated from those files, not typed. The generator verifies each file's SHA-256, recomputes every accuracy and Wilson interval, reruns every McNemar test and bootstrap from the aligned vectors, and refuses to write a macro if a stored value differs.

3.4 The information budget

The comparison is only interpretable if we know how much information each system receives. Azurite's largest context is 5,118 bytes: all 24 examples fit, but the 77 definitions do not. The public Jev harness serializes a JSON payload with the message, the labelled examples and the 77 definitions in a single choice question. We reconstruct that payload from the harness's published description and category files and the stored per-row examples, using the harness's own instruction string. The definitions alone are 7,801 bytes; the full payload has a median of 10,683 bytes (range 10,009–13,524), of which 2,367 bytes are the retrieved examples. Azurite's 5,118 bytes are 47.9% of that median. The reconstruction is not an invoice and does not recover the exact serialized prompt Jev received; it establishes the order of magnitude and the presence of the definitions, which is all the argument needs.

4 Results

4.1 The pre-registered comparison

The primary arm reaches 89.03% (95% Wilson 87.87–90.08; 2,742/3,080). Jev's frozen vector is 92.40% (2,846/3,080). The observed difference is -3.38 points; the exact McNemar test gives $p = 3.99 \times 10^{-9}$; the paired bootstrap lower bound is -4.32 points. The arm is not non-inferior at the -2.0-point margin (non-inferiority: no), and the pre-registered verdict is **P0 missed**. The interval does not contain the bar.

This is the result the paper is about, and it is a loss. The contamination audit in §3 changes its interpretation, not its verdict: the frozen 89.03% includes 255 rows that retrieve test text, while the decontaminated rerun scores 89.06% and the unaffected rows 88.88%. The loss is not an artefact of the overlap, but the frozen number is not clean. The remaining sections ask why, and which part of the loss is a head problem rather than an information problem.

4.2 Mean pooling is the immediate cause

The old head gives a clean dose-response curve at its native window. With the message only it reaches 88.38% (87.20–89.46). With the examples that fit in 1,020 bytes—11.9 on average, range 2–19—it falls to 59.12% (57.38–60.85), a loss of -33.28 points. At 5,120 bytes with all 24 examples it reaches 22.44% (21.00–23.94), -69.97

Table 1: The pre-registered comparison on all 3,080 official Banking77 test rows. *Old* is the mean-pool head; *query* is the message-query attention head. The byte budget names the context window; *examples* is the mean number of complete retrieved examples that fit (range in the text). The old-head 5,120 arm is the matched-test packing of all 24 examples; its native control is the query-evaluation packing, which averages 11.9 examples. The Jev row is the cited public run, not a rerun; its Bytes cell is the reconstructed JSON prompt size, not a model context window. Wilson intervals are 95%; Δ and exact two-sided McNemar p are against Jev's frozen per-row correctness vector. The 5,120 query head is a joint encoder/head fine-tune, not a length-only adaptation. Every cell is generated from `results/jev_query`, `results/jev_query_5120` or `results/jev_matched`.

Arm	Ex.	Bytes	n	Correct	Acc. (%)	95% Wilson	Δ vs Jev (pp)	McNemar p
Old mean-pool, message only	0.0	54	3,080	2,722	88.38	87.20–89.46	-4.03	8.49×10^{-12}
Old mean-pool, examples that fit	11.9	978	3,080	1,821	59.12	57.38–60.85	-33.28	2.96×10^{-231}
Old mean-pool, all 24 at 5,120	24.0	1,987	3,080	691	22.44	21.00–23.94	-69.97	$< 10^{-300}$
Query attention, message only	0.0	54	3,080	1,880	61.04	59.30–62.75	-31.36	1.82×10^{-226}
Query attention, examples that fit	11.9	978	3,080	2,742	89.03	87.87–90.08	-3.38	3.99×10^{-9}
Query attention, 5,120, message only	0.0	54	3,080	1,902	61.75	60.02–63.45	-30.65	1.62×10^{-232}
Query attention, 5,120, all 24	24.0	1,988	3,080	2,683	87.11	85.88–88.25	-5.29	5.01×10^{-18}
Jev 1.13, cited	24	10,683	3,080	2,846	92.40	91.41–93.29	—	—

points below Jev. The last arm changes both examples and length, so it is not a clean examples-only manipulation; the native-window arm is. Even there, the examples-only change is -29.25 points (exact McNemar $p = 4.23 \times 10^{-188}$).

The query head isolates the read-out. On the identical 1,020-byte contexts, it reaches 89.03% against the old head's 59.12%, a gain of +29.90 points (exact McNemar $p = 3.62 \times 10^{-212}$; bootstrap lower +28.41). The only difference between those two arms is the pooling operation and the head trained with it; the rows, the examples, the encoder initialisation, the training rows and the evaluation rows are the same. The pooling diagnosis is therefore confirmed at the level of the head: the examples were not unreadable, they were being averaged away.

The prediction concentration tells the same story from the output side (Figure 2). The old head at 5,120 bytes uses 64 of the 77 labels; its most frequent label takes 14.29% of rows and its three most frequent take 35.36%. The message-only old head uses all 77 labels with a top-1 share of 1.62% and a top-3 share of 4.77%. The query head with examples returns to 1.59% and 4.64% with all 77 labels. The old head is not failing because it cannot represent the labels; it is failing because the pooled context drifts toward whichever intents dominate the retrieved examples.

4.3 The fix does not close the gap

The query head with examples is 89.03%, which is 3.38 points below Jev. It is also only +0.65 points above the old head's message-only arm, and that

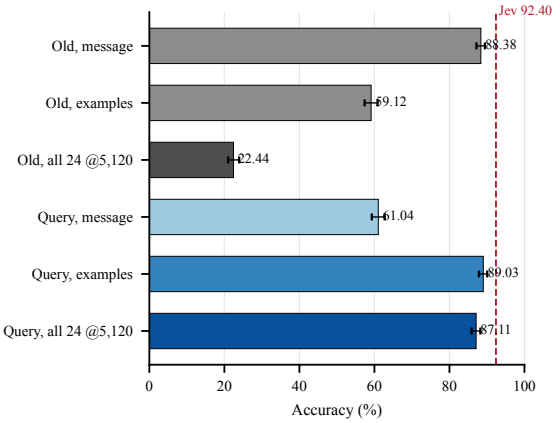


Figure 1: Accuracy on all 3,080 official Banking77 rows with 95% Wilson intervals. The dashed line is Jev's frozen 92.40%. The old head loses accuracy monotonically as examples are added; the query head recovers the native-window loss but does not beat the message-only old head. The 5,120-byte query arm is a joint fine-tune, not a length-only adaptation.

difference is not significant ($p = 2.60 \times 10^{-1}$; bootstrap lower -0.26). The point estimate is higher, but the data do not distinguish it from a head that reads only the message. At this budget the retrieved examples recover what mean pooling destroyed and buy almost nothing beyond the message itself.

The no-example query arm is 61.04%, but it is out of distribution: the query head was trained only on contexts containing examples. It is a control for the example context, not a fair no-example baseline. With that caveat, the 5,120-byte head clearly learned to use the examples—it is +25.36

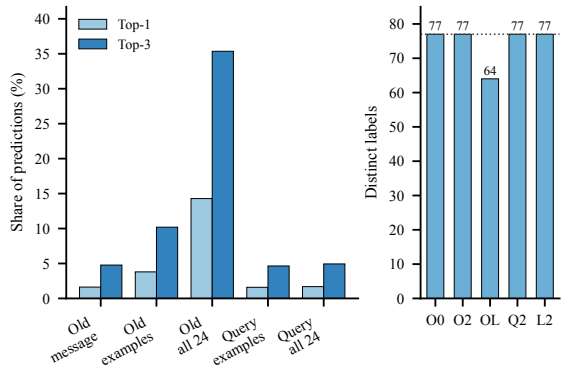


Figure 2: Prediction concentration from the stored per-row predictions. Left: share of rows assigned to the single most frequent label (top-1) and to the three most frequent labels (top-3). Right: number of distinct labels used out of 77. The old head at 5,120 bytes collapses onto a subset of labels; the query head restores the full label range. O0, O2, OL, Q2 and L2 are the old message-only, old examples, old all-24, query examples and 5,120-byte query all-24 arms.

points above its own message-only control—but the net effect against the message-only old head is still negative.

4.4 All the examples fit, and accuracy falls

The 5,120-byte arm removes the native-window example constraint. All 24 examples fit (mean 24.0, range 24–24), the mean context is 1,988 bytes, and the head is trained jointly at the longer length for 3,000 steps. It reaches 87.11% (95% Wilson 85.88–88.25), -5.29 points below Jev. Against the native-window query head it loses -1.92 points (exact McNemar $p = 8.75 \times 10^{-5}$; bootstrap lower -2.69); against the old message-only head it loses -1.27 points ($p = 3.83 \times 10^{-2}$; bootstrap lower -2.27). It does beat the old head on the same packed contexts by +27.99 points, so the query pool is still doing its job. The problem is that the longer, example-rich context is not a better decision context for this schedule.

The training/test separation bounds the claim. On the 4,000 training rows the final head reaches 96.62% (3,865/4,000) with a mean cross-entropy of 0.1321; the test accuracy is 87.11%. The test-minus-train gap is -9.51 points, the direction expected when capacity is large relative to the training set (Zhang et al., 2017). No early stopping or checkpoint selection was used, so this is the separation of the step-3,000 head, not a selected best case. A head that fits 96.62% of its training rows and still misses the bar is evidence about what it can extract from the context, not evidence that a longer fine-tune would close the gap.

4.5 Where the rows move

Table 2 replaces the aggregate accuracy with the row-level movement behind it. The query head with examples wins more rows than it loses against the old message-only head, but the net is small and the paired test is not significant ($p = 2.60 \times 10^{-1}$). Against the old head on the same contexts it wins most of the discordant rows, so the pooling recovery is broad rather than a few rescued outliers. The 5,120-byte arm loses more rows than it wins against both the native query head and the old message-only head. Moving to the longer context is not a wash: the losses are concentrated in the example-rich conditions, exactly where the longer window was supposed to help.

4.6 The example budget is not the explanation

The natural hypothesis after the native-window result is that the model simply did not have room for the examples: 1,020 bytes hold only 11.9 of 24, on average. The 5,120-byte arm tests that hypothesis and rejects it as a complete explanation. All 24 examples fit, and accuracy is lower than the native-window query head and lower than the message-only old head. More examples, in this schedule, are negative transfer.

Two things remain confounded, and we state them rather than choosing one. The 5,120-byte arm is a joint fine-tune, so length and head training move together; it cannot separate "the encoder was trained at 5,120 bytes" from "the head was trained on longer contexts". And the arm still lacks the 77 category definitions. The reconstructed comparator prompt is 10,683 bytes in the median; the definitions are 7,801 of those bytes, and the retrieved examples are 2,367 bytes. Our largest context is 47.9% of the median prompt. Even if every example is present, the information budget is not matched. The comparison can reject the examples-only explanation; it cannot attribute the remaining gap to architecture alone.

This is the information-budget bound the experiment supports. Adding examples from 11.9 to 24 does not add accuracy; the largest context effect we measure is +0.65 points, while the gap to Jev is 5.29 points. A definitions-only intervention at 5,120 bytes would need to recover several times the largest context effect measured here, and it

Table 2: Exact McNemar transition counts between selected arms. "Wins" are rows the left arm gets right and the right arm gets wrong; "losses" are the reverse. Positive net means the left arm is correct on more rows. Δ is the accuracy difference in points. Every count is recomputed from the stored per-row vectors; the table is generated with `paper/gen_paper.py`.

Comparison	Wins	Losses	Net	Δ (pp)	McNemar p
Query ex. vs old msg.	152	132	+20	+0.65	2.60×10^{-1}
Query ex. vs old ex.	986	65	+921	+29.90	3.62×10^{-212}
Long all 24 vs query ex.	81	140	-59	-1.92	8.75×10^{-5}
Long all 24 vs old msg.	149	188	-39	-1.27	3.83×10^{-2}
Old ex. vs old msg.	99	1,000	-901	-29.25	4.23×10^{-188}
Long all 24 vs long msg.	847	66	+781	+25.36	1.27×10^{-173}

would still be a different information budget from the comparator's 10,683-byte prompt. That intervention is the next pre-registered question, not a result of this paper.

5 Discussion

Constraint is not accuracy. A decider's guarantee is that the answer is in the candidate set. That guarantee is real and worth having, but it says nothing about whether the selected candidate is correct. Our result makes the distinction concrete: the best arm never emits an out-of-bank label and still misses 3.38 points of accuracy. Reports that quote only the constraint, or only a single-draw accuracy, do not support the comparative claim.

Retrieval is not free. Adding retrieved examples is usually treated as adding information. For a mean-pooled encoder it also adds noise to the representation: the message becomes a small fraction of the pooled vector. The head control shows that this is not a data problem—the query head reads the same contexts and recovers +29.90 points—but it is also not a complete fix, because the recovered accuracy is still only the message-only accuracy. Retrieval should be evaluated as a change to the read-out, not only as a change to the input.

Context length can be negative transfer. The 5,120-byte arm is the clearest negative result in the paper. All examples fit, the head is trained at the longer length, and accuracy falls relative to both the native query head and the message-only old head. Longer contexts are not automatically better; they change the distribution of the pooled representation and the effective position of the message. This is consistent with the long-context literature's warning that position and retrieval quality matter, not just window size (Liu et al., 2024; Hsieh et al., 2024), though our scale is much smaller.

What a decider comparison should report. At minimum: the exact row set; the exact examples or retrieval procedure; the category definitions or other prompt content; the context budget in bytes or tokens; the head and pooling operation; the training rows and whether they overlap the test rows; the number of seeds; and paired tests against the comparator's per-row correctness, not just two independent intervals. Our records satisfy those requirements, and the missing definitions are stated as a limitation rather than hidden.

What would change the verdict. Three interventions are worth distinguishing, and none of them is a repair of the current record. A definitions-only arm at 5,120 bytes would test whether the missing 77 definitions, rather than the examples or the pooling, account for the gap; it needs its own pre-registration because it changes the information budget. A length-only arm with a separate masked-LM adaptation would separate length from head training, which the joint schedule in this paper cannot do. A seed study at the same budget would tell us whether the +0.65 point difference is stable, though the paired interval already says it is not distinguishable from zero at this sample size. If a definitions-only arm also misses, the honest conclusion is that this model class and budget are short of the bar, not that another head will recover it.

Limitations. The experiment is one model size, one task, one seed and one head family per arm; scaling is jointly determined by parameters, data and compute, and a single rung cannot identify that surface (Kaplan et al., 2020; Hoffmann et al., 2022). We do not claim a seed distribution. The old-head 5,120-byte arm and the query-head 5,120-byte arm come from different evaluations with different packing; the table marks the dis-

tinction, and the native-window comparison is the clean one. The 5,120-byte arm is a joint fine-tune, so length and head training are confounded. The Jev vector is third-party; we freeze it rather than rerun the service, and a wrong Jev bit would change the paired tests. The prompt-budget reconstruction is a reconstruction, not a capture of the exact vendor payload. The encoder is text-only and byte-level; nothing here tests multimodality, and 5,120 bytes is not a long-context regime. Finally, the category definitions are absent from every Azurite arm, so the comparison is deliberately unequal in Jev's favour and cannot support a claim of architectural superiority.

6 Conclusion

We ran the comparison a decider claim needs: the full official Banking77 test set, the exact retrieved examples, the frozen Jev correctness vector, a pre-registered decision rule, and a head change that isolates the read-out. The best arm reaches 89.03%, 3.38 points below Jev, and misses the bar. The immediate cause is mean pooling: examples dilute the message, and a message-query attention pool recovers +29.90 points on identical contexts. The fix is not sufficient. The query head is statistically indistinguishable from the old head's message-only arm, and a 5,120-byte joint fine-tune with all 24 examples is worse than both the native query head and the message-only head. The comparator's JSON prompt is 2.09× our largest context and contains the definitions we withhold, so the experiment bounds the claim to a smaller information budget. The honest conclusion is not that deciders fail; it is that this decoder, at this budget, with this read-out, does not meet the bar, and that the next lever is information parity rather than another pooling variant.

Reproducibility

The generator `paper/gen_paper.py` reads `results/jev_query/eval/summary.json`, `results/jev_query_5120/eval/summary.json` and `results/jev_matched/summary.json`, verifies every per-example SHA-256, recomputes every statistic, and writes `paper/generated/macros.tex`, the tables, the figures and `paper/RECONCILIATION.md`. The public prompt inputs are vendored under `paper/data/jev_prompt` with their own hashes and source record. Rebuilding is `python paper/gen_paper.py && cd paper && tectonic azurite.tex`.

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Table 3: Content identities for the records the paper reads. Each is the first 12 hex digits of a SHA-256; the full values and the per-example hashes are in the stored summaries and `paper/RECONCILIATION.md`.

Artifact	SHA-256 (first 12)
Query pre-registration	`bde97e74f2f8`
Query inputs	`8dccd8c89886`
Query code	`95a5e0a5d11e`
Old head	`42f1ed1c8471`
Query head	`7992c97362b3`
5,120 pre-registration	`c59b5e08d916`
5,120 amendment	`7993177d769a`
5,120 init checkpoint	`5294fd685615`
5,120 head	`e7455fc462e9`
5,120 code	`0800b2aad102`

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A Statistical definitions

For an arm with k correct rows out of n, the 95% Wilson interval is

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \qquad (1)$$

with $z = 1.959963984540054$ and $\hat{p} = k/n$ (Wilson, 1927). For two aligned correctness vectors, the exact McNemar test counts the discordant pairs b (left correct, right wrong) and c (left wrong, right correct) and reports the two-sided binomial tail at $p = 1/2$ (McNemar, 1947). The paired bootstrap resamples rows with replacement, computes the mean difference in each of 10,000 draws and reports the 5th percentile; the seed is 0. The P0 rule and the non-inferiority margin are fixed in §3.2. All three quantities are recomputed from the stored per-row vectors by `paper/gen_paper.py`.

B Content identities and reproduction

The full SHA-256 values and per-example hashes are in `paper/RECONCILIATION.md` and the three summary JSON files. The reconciliation step also checks the shared values against the thesis macros, so the paper and the thesis cannot drift apart silently.

Records. The query evaluation is under `results/jev_query/`; the 5,120-byte evaluation under `results/jev_query_5120/`; the matched-test and the old-head 5,120-byte arm under `results/jev_matched/`.